FIFTH SEMESTER (CUFYUGP) DEGREE EXAMINATION: MODEL PAPER 1

PHYSICS / APPLIED PHYSICS

ELECTRODYNAMICS II

Module 1: Electric and Magnetic Fields in Matter

Maximum Time: 2 Hours		**Maximum Marks: 70**
Section A		
All Questions can be answered. Each Question carries 3 marks (Ceiling: 24 Marks)		
1	Write down the relation connecting polarization and electric field in linear dielectrics. What do you mean by a polar dielectric?	**3**
2	Give the relation connecting permittivity and susceptibility in linear dielectrics. Explain the terms involved. How is relative permittivity connected to susceptibility?	**3**
3	Explain physically why a string of head-to-tail dipoles leaves net charge at its two ends.	**3**
4	Write down Gauss's law in presence of a dielectric.	**3**
5	State the boundary condition on $D^{\perp}$ and on $\mathbf{D}^{\parallel}$ at an interface, in terms of free charge and polarization.	**3**
6	Distinguish between permeability and permittivity.	**3**
7	What is the basic difference between electric and magnetic polarizations?	**3**
8	Explain physically why a uniformly magnetized slab carries a surface current, although each atom carries only a tiny loop current.	**3**
9	Discuss the magnetic field inside matter.	**3**
10	Explain the temperature-driven phase transitions in ferromagnetic materials.	**3**
Section B		
All Questions can be answered. Each Question carries 6 marks (Ceiling: 36 Marks)		
11	Show that the energy of an ideal dipole $\vec{p}$ in an electric field $\vec{E}$ is given by $U = -\vec{p} \cdot \vec{E}$.	**6**
12	Derive the expression for capacitance of a parallel plate capacitor with partially filled dielectric.	**6**
13	Explain the hysteresis loop of ferromagnets.	**6**
14	Explain the domain model of ferromagnets.	**6**
15	Write an expression for the field of a magnetized object. Explain the terms involved.	**6**
16	A solid sphere of radius 2 cm is uniformly polarized with $P = 1.0 \times 10^{-6}$ C/m^2. Find the field inside the sphere and the dipole moment of the sphere. Take $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m.	**6**
17	A uniformly magnetized sphere of radius 5 cm has $M = 1.0 \times 10^4$ A/m. Find the magnetic field inside the sphere and its magnetic dipole moment. Take $\mu_0 = 4\pi \times 10^{-7}$ H/m.	**6**
18	A long solenoid with $n = 1000$ turns/m carries a current $I = 2$ A and is filled with a linear material of $\chi_m = 2 \times 10^{-5}$. Find $\mathbf{H}$, $\mathbf{B}$ inside, and the bound surface current density on the surface of the material.	**6**
Section C		
Answer any ONE. Each Question carries 10 marks (1x10=10 Marks)		
19	Explain the terms polarization and displacement. Derive the relation connecting them. Derive an expression for potential of polarized objects.	**10**
20	Discuss in detail Ampère's law in magnetized materials. Derive an expression for the magnetic field at any point inside and outside of a long copper rod of radius R carrying a uniformly distributed current I.	**10**